Giga-sample Data Acquisition Method for High-speed DDR5 SDRAM

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***Abstract*—We propose a data acquisition system that enables oversampling of signals with an external high-speed clock, to overcome the unreliability of the existing data acquisition method that uses a DRAM memory clock, when applied to data rate fifth-generation synchronous dynamic random-access memory [1]. We present a simple noise-modeling method and signal-recovery algorithm with a noise range where our algorithm behaves reliably at error rate $< 10^{-6}$. In addition, we propose an entire data acquisition system architecture that uses a Xilinx FPGA VCU118 board.**

***Index Terms*—Oversampling, FPGA, noise modeling, signal recovery**

I. Introduction

Current CPUs and GPUs have multiple cores that operates at several-gigahertz clock frequencies, and memory technology is being modified accordingly to process large amounts of data. However, if the data-transfer speed fails to keep up with the computing speed, data-transfer limits the speed of the system. To maintain high data-transfer speed, the data bandwidth of the next generation memory, DDR5 SDRAM [1], has been increased to 6.4 Gbps, which is double the rate of the previous generation, and the oprating voltage has been reduced to 1.1 V. Increased data-transfer speed leads to increased current usage, which results in various noises, which hinder use of the existing data acquisition method that uses the DRAM memory clock.

A data acquisition system can instead exploit an external high-speed clock. The system provides oversampled data received by a high-speed clock that operates independently of the DRAM system clock. Using high-speed serial I/O such as Xilinx GTY Transceiver that has a data rate up to 32.75 Gbps [2], the oversampled data can be restored to actual data by a signal-recovering process even in noisy environments.

In this paper, we simply model how noise is represented and propose a recovery algorithm with a guaranteed noise range in which the proposed algorithm operates reliably. We also propose an entire data acquisition system architecture that uses the proposed recovery algorithm. The main contributions of our work are:

- We model digital signals and propose a signal recovery algorithm that can work reliably in a noisy environment.
- We investigate the trade-off between tolerance to jitter and tolerance to oversampling variation for an algorithm that operate at error rate $< 10^{-6}$.
- We propose an entire data acquisition system architecture that uses the proposed recovery algorithm and a Xilinx FPGA VCU118 board.

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TABLE I
COMPARISON OF DDR5 SDRAM AND DDR4 SDRAM [1].

	DDR5 SDRAM	DDR4 SDRAM
Max data rate	6.4 Gbps	3.2 Gbps
Bank (per group)	4	4
Bank group	8 (x4, x8), 4 (x16)	4 (x4, x8), 2 (x16)
Burst-length	BL16	BL8
Operation voltage	1.1 V	1.2 V
Max die density	64 Gbit	16 Gbit

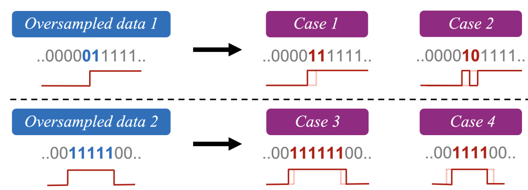

Fig. 1. Signal noise modeling of four cases.

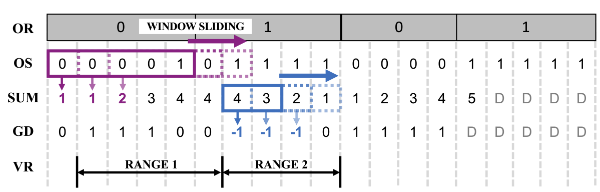

Fig. 2. Clock and data recovery algorithm flow. OR: original signal; OS: oversampled signal; SUM: summation; GD: gradient; VR: voting range.

II. Proposed Method

A. Noise Modeling

Transitions generate various kinds of noise [3], which can be represented digitally (Fig. 1). This unstable situation causes $Case$ 1 and $Case$ 2, which are jitter-cause situations. In $Case$ 1, the last value is converted to the next signal's value. In $Case$ 2, two adjacent values are exchanged at the transition point. $Case$ 3 and $Case$ 4 are oversampling-noise situations in which the signal is oversampled with a different ratio from the target ratio.

B. Clock and Data Recovery Algorithm

Oversampled data includes four cases of signal noises. To recover the original signal (OR), the effect of jitter must be removed. $Average\ Phase\ Picking$ (APP) [4] is an algorithm that uses a fixed-length interval to reduce the impact of each jitter. In our proposed clock and data recovery algorithm (Fig. 3), we define the momentum as the number of times that the same gradient (GD) comes out, and apply it to APP algorithm. First, to reduce the effect

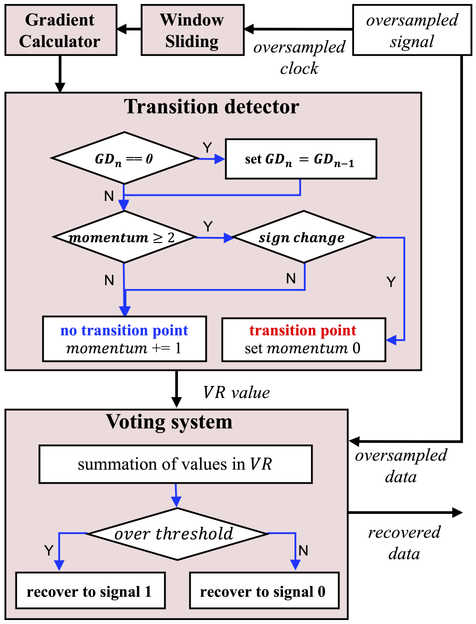

Fig. 3. Clock and data recovery algorithm flow.

of each jitter element, the values of the oversampled signal (OS) is summed by sliding the window; this process changes the OS to a summation (SUM). Second, the GD, the difference between two adjacent SUM, is obtained to determine the phase-transition point. By exploiting the property that the clock is always transitions, point when the sign of the GD changes is selected as the transition point.

Two decision criteria are considered. First, when GD = 0, then GD is set to the previous value. Second, if the momentum does not exceed two, the change in GD is ignored. These two conditions allow selection of an appropriate voting range (VR) for reliable clock and data recovery in complex jitter-cause and oversampling-noise situations.

Finally, the voting system (Fig. 3) determines the recovered signal by using the sum of the OS in VR. In the voting system, the threshold value is selected by the length of the VR. For example, in RANGE 1, which has five elements, the signal is recovered to 1 if the sum of the values is $\geq$3. In RANGE 2, which has four elements, the signal is recovered to 1 if the sum of the values is $\geq$2. The proposed algorithm recovers the clock signal first and then uses the same VR to recover the data signal.

C. Data acquisition system block diagram

The data acquisition system (Fig. 4) uses a Xilinx FPGA VCU118 board [5] to deliver the command signals between DDR5 SDRAM and HOST to the GTY Transceiver, which generates oversampled data by using the external high-speed clock, and stores them in a data buffer. The proposed recovery hardware restores the original data by using a clock and data recovery algorithm. A MicroBlaze, Xilinx FPGA soft processor core, can monitor and validate the workload between DDR5 SDRAM and HOST.

III. Experiment Result

When Bit-Error-Rate (BER) is E^{-9}, JEDEC recommends testing to 99.5% confidence levels with 5.298 BER [1]. That is, the noise rate required for behavior validation of DDR5 SDRAM is very low. However, in this experiment, we assume an extremely noisy environment, so we allowed the rate of occurrences in each case up to 5%.

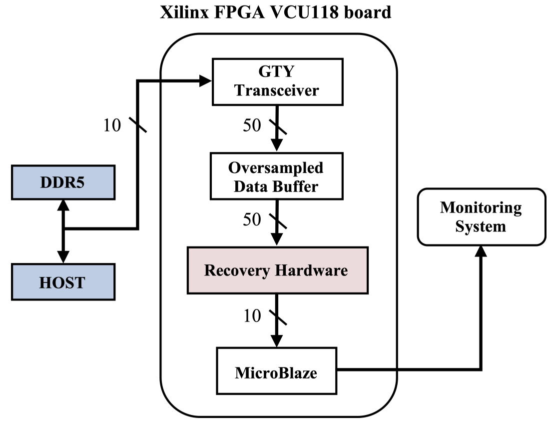

Fig. 4. Proposed data acquisition system for DDR5 SDRAM.

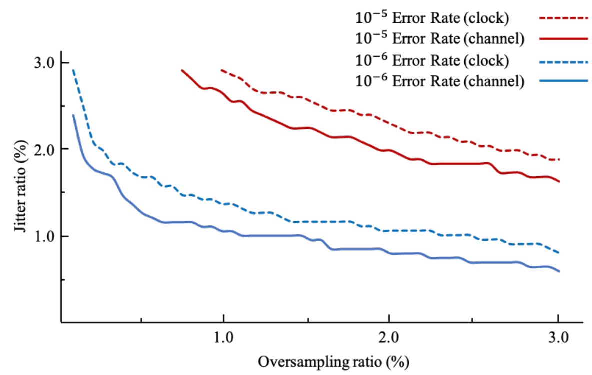

Fig. 5. Clock and data recovery algorithm result for 100,000 PRBS data. Red and blue line represent the Pareto curve that recovers original signal under 10^{-3} and 10^{-4} error rate, respectively.

The clock and data recovery algorithm was conducted for 5,000,000 PRBS data in jitter and oversampling variation environment. CASE 1 and CASE 2 were assumed to occur in equal ratio. CASE 3 and CASE 4 were also assumed to occur in equal ratio to make the signal to be oversampled by an average of five times. The recovery algorithm worked reliably in environment in which jitter and oversampling noise are present (Fig. 5).

IV. Conclusion

We proposed a data-acquisition method that uses an external high-speed clock for validation of DDR5 SDRAM operation. We also propose a signal-recovery algorithm that is reliable in extremely noisy situations. Additionally, we find Pareto curve that determine the range of jitter and oversampling noise in which the proposed algorithm operates reliably.

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